

**Ubuntu Innovations
(Pty) Ltd**



IT Infrastructure Documentation Report

Prepared by :
Khusela Ikamva Tech

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1. Business Requirements Analysis

1.1 Company Overview

Ubuntu Innovations (Pty) Ltd is a South African technology SME currently relocating to a new office. The company employs 25 full-time staff across five departments and requires a complete IT infrastructure to be designed, deployed, and documented from scratch to support day-to-day business operations.

As the Junior IT Support Specialist team, this report forms the first deliverable of a four-week build plan. It identifies who works at the company, what they do, what technology they need, and what risks need to be planned for.

Department	Headcount	Primary Function
Executives	3	Strategic leadership, board reporting, company governance
Finance	4	Accounting, payroll, budgeting, online banking
HR	4	Recruitment, employee records, compliance, training
Sales & Marketing	7	Client acquisition, campaigns, CRM, social media
Software Development	7	Product development, version control, cloud deployments
TOTAL	25	Full-spectrum SME operations

1.2 Departmental IT Requirements

1.2.1 Executives (3 staff)

The Executive team is responsible for strategic leadership, investor relations, and board-level reporting. Their IT requirements focus on mobility, secure access, and reliable communication tools.

Software needed:

- Microsoft 365 — Outlook, Word, Excel, PowerPoint, Teams
- Zoom or Microsoft Teams for remote meetings and board calls
- SharePoint or OneDrive for secure cloud document access
- Antivirus and BitLocker device encryption on all laptops

Hardware needed:

- 3 × premium business laptops (Dell Latitude or Lenovo ThinkPad)
- Docking stations and dual monitors for in-office use
- Smartphones enrolled in Mobile Device Management (MDM)

1.2.2 Finance Department (4 staff)

The Finance team handles all financial transactions, payroll, SARS tax submissions, and online banking. They work with highly sensitive data and require secure, regulated access to financial systems.

Software needed:

- Sage Accounting or Xero for bookkeeping and financial reporting
- Microsoft Excel (advanced) for budgets and reconciliations
- Secure browser access to online banking portals
- eFiling access (SARS) for VAT, PAYE, and income tax submissions

- Adobe Acrobat for financial statements

Hardware needed:

- 4 × business desktops or laptops with dual monitors
- Dedicated network printer/scanner for physical financial documents
- UPS at every Finance workstation

1.2.3 Human Resources Department (4 staff)

The HR team manages recruitment, employee records, leave tracking, and compliance under the BCEA. They handle personally identifiable information and must comply with POPIA.

Software needed:

- HR management system: SimplePay, Sage HR, or BambooHR
- Microsoft 365 for document creation, email, and internal communication
- Secure digital filing system for employee contracts and ID documents
- Video conferencing for remote interviews (Teams or Zoom)

Hardware needed:

- 4 × business laptops or desktops
- Access to the shared office printer/scanner

1.2.4 Sales & Marketing Department (7 staff)

The Sales and Marketing team drives business growth through client acquisition, digital marketing campaigns, and CRM. They are the highest-traffic users of cloud tools and external web platforms.

Software needed:

- CRM platform: HubSpot, Salesforce, or Zoho CRM
- Email marketing: Mailchimp or HubSpot for campaigns and newsletters
- Social media management: Hootsuite or Buffer
- Design tools: Canva Pro or Adobe Express
- Microsoft 365 for proposals, presentations, and email

Hardware needed:

- 7 × laptops — mix of mid-range and higher-spec for designers
- Large monitors for design and presentation work
- Access to colour printer for marketing collateral

1.2.5 Software Development Department (7 staff)

The Software Development team builds, tests, and deploys Ubuntu Innovations' core products. They are the most technically demanding users, requiring high-spec hardware and secure access to cloud and version control environments.

Software needed:

- Version control: GitHub or GitLab for source code management
- Development IDEs: Visual Studio Code, IntelliJ IDEA, or PyCharm
- Containerisation: Docker Desktop and/or Kubernetes tools
- Cloud platforms: AWS, Azure, or Google Cloud Console
- Project management: Jira or GitHub Projects for sprint tracking

Hardware needed:

- 7 × high-spec workstations — min. 32 GB RAM, 1 TB NVMe SSD, multi-core CPU
- Dual or triple monitor setups per developer

- Dedicated VLAN network segment separated from general staff

1.3 Shared IT Infrastructure Requirements

- High-speed fibre internet (minimum 100 Mbps symmetric) with 4G LTE failover router
- Managed network switch and enterprise Wi-Fi access points for all 25 users
- Windows Server or cloud-based Active Directory for centralised user management
- Network-attached storage (NAS) for shared drives and local backups
- Company-wide Microsoft 365 Business Standard licences for all staff
- Centralised antivirus management console (ESET Endpoint Security or Bitdefender GravityZone)
- UPS units in the server room and at Finance workstations

1.4 Risk Analysis

Risk / Threat	Mitigation Strategy	Severity
Power Outages (load-shedding)	UPS devices, surge protection, generator backup	HIGH
Internet Connectivity Loss	Dual ISP lines or LTE failover router	HIGH
Data Loss / Ransomware	Daily offsite backups, antivirus, staff training	CRITICAL
Unauthorised Access	Strong passwords, MFA, role-based permissions	HIGH
Hardware Failure	Warranty support, spare parts, cloud storage	MEDIUM
Staff IT Misuse	Acceptable use policy, content filtering	MEDIUM
Software Licensing Issues	Asset register, licensed software only	MEDIUM

1.5 Conclusion & Next Steps

Ubuntu Innovations (Pty) Ltd has clear and well-defined IT requirements across all five departments. The infrastructure must support 25 employees reliably, securely, and cost-effectively — from executive mobility through to developer workstations.

- Week 1 (current) — Procure hardware, set up network equipment, configure workstations
- Week 2 — Install operating systems, configure user accounts, set up shared drives
- Week 3 — Implement security policies, configure backups, deploy antivirus
- Week 4 — Final testing, staff onboarding, and handover documentation

2. Hardware Inventory

Total line items: 27
Total units: 209
Estimated total cost (excl. VAT): R 1,174,550
Prepared: May 2026

2.1 Computing Equipment

Item	Description	Specifications	Dept	Qty	Unit (ZAR)	Total (ZAR)	Priority
COMP-01	Executive Laptop	Intel i7-1355U, 16 GB DDR5, 512 GB NVMe, Win 11 Pro	Exec	3	28,500	85,500	HIGH
COMP-02	Finance & HR Desktop	Intel i5-13400, 16 GB DDR4, 512 GB NVMe, Win 11 Pro	Fin, HR	8	14,500	116,000	HIGH
COMP-03	Sales & Mktg Laptop	Intel i5-1335U, 16 GB DDR4, 512 GB SSD, Win 11 Pro	Sales & Mktg	7	18,000	126,000	HIGH
COMP-04	Developer Workstation	i7-13700K, 32 GB DDR5, 1 TB NVMe, RTX 3060	Dev	7	38,000	266,000	HIGH

2.2 Monitors

Item	Description	Specifications	Dept	Qty	Unit (ZAR)	Total (ZAR)	Priority
MON-05	24" Full HD Monitor	1920x1080, IPS, 75 Hz, HDMI+DP	Fin, HR, Exec	16	4,200	67,200	HIGH
MON-06	27" QHD Monitor	2560x1440, IPS, 144 Hz, USB-C	Dev, Sales	10	7,500	75,000	MEDIUM

2.3 Networking Equipment

Item	Description	Specifications	Dept	Qty	Unit (ZAR)	Total (ZAR)	Priority
NET-07	Core Managed Switch	48-port Gigabit, PoE+, VLAN, QoS, rack-mount	Srv Room	1	18,000	18,000	CRITICAL
NET-08	Wi-Fi Access Point	Wi-Fi 6, dual-band, WPA3, PoE-powered	All Depts	4	5,500	22,000	HIGH
NET-09	Fibre Router/Firewall	1 Gbps WAN, VPN, stateful firewall, dual WAN	Srv Room	1	12,000	12,000	CRITICAL
NET-10	LTE Failover Router	4G LTE Cat-12, dual SIM, load balancing	ISP Backup	1	6,500	6,500	HIGH
NET-11	Cat6A Cabling (per pt)	30 data points, RJ45, patch panels	All Depts	30	850	25,500	HIGH

2.4 Servers & Storage

Item	Description	Specifications	Dept	Qty	Unit (ZAR)	Total (ZAR)	Priority
SRV-12	File & AD Server	Xeon E-2336, 32 GB ECC, RAID 1+5, Win Server 2022	Srv Room	1	55,000	55,000	CRITICAL
SRV-13	NAS Device	4-bay, 4x4 TB IronWolf, RAID 5, SMB/NFS	All	1	22,000	22,000	HIGH
SRV-14	Server Rack 12U	Lockable, vented, cable management	Srv Room	1	8,500	8,500	MEDIUM

2.5 Power Protection

Item	Description	Specifications	Dept	Qty	Unit (ZAR)	Total (ZAR)	Priority
PWR-15	Server Room UPS 3000VA	Pure sine wave, 8x IEC, SNMP, LCD display	Srv Room	1	18,000	18,000	CRITICAL
PWR-16	Workstation UPS 1200VA	Simulated sine, 4x outlet, USB shutdown	Finance, HR	6	2,800	16,800	HIGH
PWR-17	Surge Protector Strip	6-outlet, 2500 joule, indicator light	All Depts	15	380	5,700	MEDIUM

2.6 Peripherals, Printing & Security

Item	Description	Specifications	Dept	Qty	Unit (ZAR)	Total (ZAR)	Priority
PER-18	Wireless Keyboard & Mouse	Full-size, USB nano receiver, 12-month battery	All Depts	25	650	16,250	MEDIUM
PER-19	USB-C Docking Station	2x HDMI, 4x USB-A, Gigabit LAN, 100W PD	Exec, Sales	10	2,200	22,000	MEDIUM
PER-20	USB Wired Headset	Noise-cancelling mic, USB-A, Teams certified	All Depts	25	900	22,500	MEDIUM
PER-21	Full HD Webcam	1080p, auto-focus, privacy shutter	Exec, Sales	14	1,200	16,800	MEDIUM
PRT-22	Multifunction Laser MFP	A4/A3 colour laser, duplex, LAN+Wi-Fi, 500-sheet	Fin, HR, Sales	2	18,000	36,000	MEDIUM
PRT-23	Label Printer	Thermal, 300 dpi, USB+LAN	IT / HR	1	2,800	2,800	LOW
SEC-24	IP Camera System	4MP, 8-ch NVR, 2 TB HDD, PoE, night vision	Office	1	14,500	14,500	HIGH
SEC-25	Smart Door Access Control	RFID + PIN, 25x cards, audit log, 12V DC	Srv Room	1	6,500	6,500	HIGH
TEL-26	VoIP Desk Phone	SIP, 2-line, HD voice, PoE, Gigabit	Exec, Fin	10	2,500	25,000	MEDIUM
MOB-27	Company Android Phone	Android 14, 128 GB, 5G, enrolled in Intune MDM	Exec, Sales	7	9,500	66,500	MEDIUM

2.7 Hardware Budget Summary by Category

Category	Items	Units	Cost (ZAR)	% of Budget	Priority
Computing	4	25	593,500	50.5%	CRITICAL / HIGH
Monitors	2	26	142,200	12.1%	HIGH / MEDIUM
Networking	5	37	84,000	7.2%	CRITICAL / HIGH
Server & Storage	3	3	85,500	7.3%	CRITICAL / HIGH
Power Protection	3	22	40,500	3.4%	CRITICAL / HIGH
Peripherals	4	74	77,550	6.6%	MEDIUM
Printing	2	3	38,800	3.3%	MEDIUM / LOW
Security	2	2	21,000	1.8%	HIGH
Telephony	1	10	25,000	2.1%	MEDIUM
Mobile / MDM	1	7	66,500	5.7%	MEDIUM
TOTAL	27	209	1,174,550	100%	—

All prices are estimated retail/tender prices in South African Rand (ZAR) as at May 2026. Actual quotes may vary by 5–15%. VAT (15%) is excluded from all figures.

3. Software Inventory

Total licences: 232
Estimated annual software cost (excl. VAT): R 638,440
Last updated: May 2026

3.1 Human Resources

Software	Category	License Type	Licences	Cost (ZAR/yr)	Notes
Microsoft Office 365	Productivity Suite	Subscription	10	18,000	Word, Excel, Outlook, Teams
PaySpace Payroll	Payroll	Subscription	5	24,000	South African payroll & HR
BambooHR	HR Management	Subscription	3	15,000	Employee records & onboarding
Adobe Acrobat	Document Management	Subscription	5	9,000	PDF creation & editing

3.2 IT Department

Software	Category	License Type	Licences	Cost (ZAR/yr)	Notes
Microsoft 365 Admin	Administration	Included	2	0	Tenant management
ManageEngine ServiceDesk	Helpdesk / ITSM	Subscription	5	32,000	IT ticketing system
SolarWinds RMM	Network Monitoring	Subscription	3	28,000	Remote monitoring & management
Veeam Backup	Backup & Recovery	Perpetual	2	21,000	Server backup solution

3.3 Software Development

Software	Category	License Type	Licences	Cost (ZAR/yr)	Notes
Visual Studio Code	IDE / Code Editor	Free	15	0	Open source — no cost
JetBrains Suite	IDE	Subscription	10	45,000	IntelliJ, PyCharm, WebStorm
Git / GitHub Enterprise	Version Control	Subscription	15	38,000	Source code management
Docker Desktop	Containerisation	Subscription	15	22,000	Container runtime & tooling
Postman	API Testing	Subscription	8	9,600	API development & testing
Jira Software	Project Management	Subscription	15	18,000	Agile project tracking

3.4 Finance

Software	Category	License Type	Licences	Cost (ZAR/yr)	Notes
Microsoft Office 365	Productivity Suite	Subscription	8	14,400	Word, Excel, Outlook, Teams
Sage 300 ERP	Accounting / ERP	Perpetual	5	85,000	Financial management & reporting
Power BI Pro	Business Intelligence	Subscription	5	12,000	Financial dashboards & reports
Adobe Acrobat	Document Management	Subscription	4	7,200	PDF creation & editing

3.5 Sales & Marketing

Software	Category	License Type	Licences	Cost (ZAR/yr)	Notes
Microsoft Office 365	Productivity Suite	Subscription	12	21,600	Word, Excel, Outlook, Teams
Salesforce CRM	CRM	Subscription	12	72,000	Customer relationship management
HubSpot Marketing Hub	Marketing Automation	Subscription	4	36,000	Email campaigns & analytics
Canva for Teams	Design	Subscription	8	9,600	Marketing collateral & graphics
Zoom	Video Conferencing	Subscription	12	8,640	Client calls & webinars

3.6 Operations

Software	Category	License Type	Licences	Cost (ZAR/yr)	Notes
Microsoft Office 365	Productivity Suite	Subscription	8	14,400	Word, Excel, Outlook, Teams
Asana	Project Management	Subscription	8	14,400	Task & project tracking
Slack	Communication	Subscription	10	9,600	Internal team messaging
Microsoft Visio	Diagramming	Subscription	3	9,000	Process flow diagrams

3.7 Executive / Management

Software	Category	License Type	Licences	Cost (ZAR/yr)	Notes
Microsoft Office 365	Productivity Suite	Subscription	5	9,000	Word, Excel, Outlook, Teams
Power BI Pro	Business Intelligence	Subscription	5	12,000	Executive dashboards
DocuSign	E-Signature	Subscription	5	18,000	Digital contract signing

Software	Category	License Type	Licences	Cost (ZAR/yr)	Notes
LastPass Business	Password Management	Subscription	5	6,000	Secure credential management

3.8 Annual Software Cost Summary

Total estimated annual software spend: R 638,440 (232 licences across 7 departments). Excludes VAT, hardware, infrastructure, and installation. Open-source tools - VS Code and Git -are listed at R 0.

4. Network Topology Diagram

The diagram below shows the logical network topology for Ubuntu Innovations' new office. Internet traffic enters through the firewall appliance, passes through the core managed switch, and is distributed to four categories of endpoint devices: desktop computers, wireless access points, shared printers, and NAS storage.

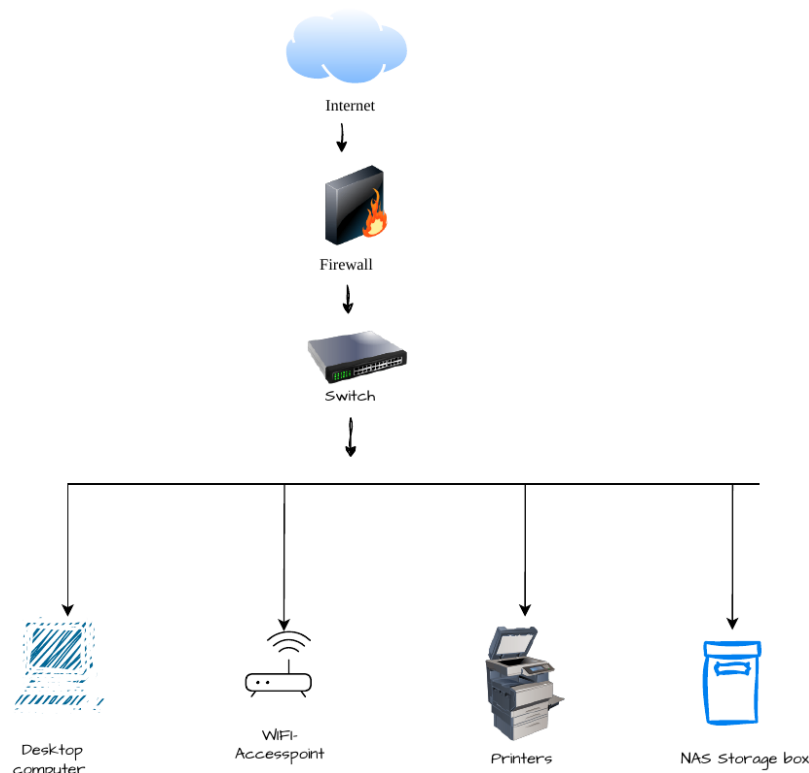


Figure 1 — Ubuntu Innovations Office Network Topology

4.1 Network Component Summary

Component	Description
Internet	External WAN via fibre ISP with 4G LTE failover
Firewall	FortiGate 60F or Ubiquiti Dream Machine — stateful inspection, VPN, dual WAN
Core Switch	Cisco SG350-48P — 48-port PoE+ managed switch with VLAN support
Desktop Computers	25 workstations and laptops across all five departments
Wi-Fi Access Points	4 x Ubiquiti UniFi U6-Pro (Wi-Fi 6) for full office wireless coverage
Printers	2 x Multifunction colour laser printers shared across Finance, HR, and Sales
NAS Storage	Synology DS923+ — 4-bay RAID 5 for shared drives and local backups

5. Troubleshooting Scenarios

This section documents step-by-step resolution procedures for the three most frequently reported IT issues across all departments. Each scenario includes a problem summary, affected systems, estimated resolution time, and a clear action plan. All IT support staff are expected to follow these procedures before escalating to Tier 2.

#	Scenario	Issue Type	Severity	Est. Resolution
1	Printer Offline	Connectivity / Hardware	Medium	15–30 min
2	Password Reset / Account Locked	Security / Access	High	5–10 min
3	Slow Computer Performance	Performance / Software	Low	30–60 min

5.1 Scenario 1 — Printer Offline

Severity: Medium
 Est. Resolution: 15–30 minutes
 Affects: All departments
 Likely Causes: Power/cable issues, network, outdated drivers, queue corruption

#	Action	Details / Instructions
1	Check Power & Cables	Confirm the printer is switched on and the power cable is secure. Check the network cable or confirm Wi-Fi signal. Look for error lights on the control panel.
2	Restart the Printer	Power off completely, wait 30 seconds, then power back on. Wait for full initialisation before proceeding.
3	Verify Network Connection	Print a configuration page (hold Info button 3 sec) and confirm the printer shows a valid IP address on the same subnet as the user's PC.
4	Clear the Print Queue	Control Panel > Devices & Printers > right-click printer > See what's printing > Cancel all. If stuck: stop Print Spooler in services.msc, delete files in C:\Windows\System32\spool\PRINTERS, restart the service.
5	Set Printer Online	Right-click printer > See what's printing > Printer menu. Uncheck 'Use Printer Offline' if ticked.
6	Reinstall / Update Driver	Right-click printer > Remove device. Download the latest driver from the manufacturer's website and reinstall, or use Windows Update > Optional updates.
7	Test Print	Right-click printer > Printer properties > Print Test Page. Confirm the page prints correctly.
8	Escalate if Unresolved	Log a ticket in ManageEngine ServiceDesk with the printer IP, model, and a screenshot of any error. Assign to Tier 2 — Network/Hardware team.

5.2 Scenario 2 — Password Reset / Account Locked

#	Action	Details / Instructions
1	Verify Identity	Before any action, confirm identity via employee ID, security question, or registered mobile number using the company's MFA system.
2	Check Account Status in AD	Open Active Directory Users & Computers (ADUC). Search for the user. Check if 'Account is locked out' is shown. Note the lockout timestamp.
3	Unlock the Account	In ADUC: right-click user > Properties > Account tab > tick 'Unlock account' > Apply. Or run in PowerShell: Unlock-ADAccount -Identity username

#	Action	Details / Instructions
4	Reset the Password	Right-click user in ADUC > Reset Password. Set a temporary password (min. 10 chars: upper + lower + number + symbol). Tick 'User must change password at next logon'.
5	Clear Cached Credentials	Go to Credential Manager (Control Panel > Windows Credentials). Remove stale domain entries to prevent old passwords triggering repeat lockouts.
6	Communicate to User	Send the temporary password via SMS or phone call — never via unencrypted email. Instruct the user to log in and change it immediately.
7	Investigate Lockout Cause	Check Windows Security Event Log (Event ID 4625, 4740) on the domain controller. Common causes: mobile email app with old password, mapped drive, or scheduled task.
8	Document the Incident	Log in ManageEngine ServiceDesk: user, time, lockout source, and resolution steps. Flag if a security breach is suspected.

5.3 Scenario 3 — Slow Computer Performance

#	Action	Details / Instructions
1	Check Resource Usage	Open Task Manager (Ctrl+Shift+Esc) > Performance tab. If CPU, RAM, or Disk is consistently above 85%, identify the top process in the Processes tab.
2	Close Unnecessary Programs	Ask the user which apps they need. Close everything else, including hidden system tray apps like Teams, OneDrive, or antivirus scanners running updates.
3	Restart the Computer	A restart clears RAM, applies pending Windows updates, and fixes many temporary slowdowns. Ensure the user saves all work first, then re-test.
4	Disable Startup Programs	Task Manager > Startup tab. Disable non-essential apps (Spotify, Skype, Steam). Restart and re-test.
5	Run Disk Cleanup	Open Disk Cleanup, select C: drive, tick all boxes including System files. Delete temp files, old updates, and Recycle Bin. Ensure at least 15% free disk space.
6	Scan for Malware	Run a full scan using Microsoft Defender or CrowdStrike. Quarantine any threats and report infections to the security team immediately.
7	Check Windows Updates	Settings > Windows Update. Install all pending updates and restart. Outdated systems run slower due to unpatched bugs.
8	Check Hardware Health	Use HWMonitor to check CPU temp (under 85 deg C under load). Run Windows Memory Diagnostic for RAM. Use CrystalDiskInfo for HDD health — flag bad sectors.
9	Recommend Hardware Upgrade	If the PC is over 5 years old, under 8 GB RAM, or running an HDD, document the specs and submit a hardware upgrade request to the IT Manager.

5.4 Escalation Policy

Tier	Who Handles	Scope	Escalate After
Tier 1	IT Helpdesk	Password resets, printer issues, basic software problems	30 minutes
Tier 2	IT Engineers	Network issues, hardware failures, complex software, AD/Server	2 hours
Tier 3	Senior IT / Vendor	Server outages, security incidents, ERP/core system failures	4 hours

**Ubuntu Innovations
(Pty) Ltd**



IT Operating Systems & User Access Report

Prepared by :
Khusela Ikamva Tech

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Executive Summary & Introduction

As **Ubuntu Innovations (Pty) Ltd.** continues to expand its operational footprint, the requirement for robust, reliable, and secure information technology infrastructure becomes paramount to maintaining operational efficiency and data integrity. This technical report outlines the strategic implementation of the company's systems administration baseline, endpoint deployment frameworks, and centralized storage architectures.

The objective of this initiative is twofold: first, to establish rigorous role-based access controls and secure data directory management via localized systems administration; and second, to modernize our enterprise data storage through a structured Network-Attached Storage (NAS) framework. By transitioning **Ubuntu Innovations (Pty) Ltd.** from legacy, flat-network environments to optimized, segmented subnets, this deployment ensures absolute data confidentiality, simplifies firewall administration, and provides a highly scalable foundation capable of supporting ongoing corporate growth.

Access Control Policy and Security Group Mapping

To maintain a secure and scalable network ecosystem, system access is governed strictly through an enterprise **User Permissions Matrix**. Rather than applying explicit security permissions to individual user accounts—an administrative practice that introduces auditing complexures during employee turnover—permissions are mapped exclusively to functional security groups.

Utilizing this structural framework, each directory operates as a secured boundary, where membership within a corresponding Active Directory (AD) security group acts as the cryptographic credential required for access. Users who are not explicitly enrolled in the designated group are systematically denied entry, ensuring absolute data isolation across the enterprise.

Who gets which key

Team	Security group	What they can get into
Management	mgmt_grp	Full access across the board. They can open and change anything.

Finance	finance_grp	Finance folder only. Everything else is closed off.
HR	hr_grp	HR folder only. Same deal — nothing outside their area.
Sales	sales_grp	Sales folder only.
Development	dev_grp	Dev folder only.

Administrative Escalation and Auditing Oversight

While executive management retains comprehensive administrative access privileges across all file system directories, day-to-day operational data management remains delegated strictly to respective department personnel. Executive-level access is maintained primarily to facilitate independent compliance audits, ensure administrative redundancy, and allow for formal directorial verification or sign-off on isolated assets during exceptional operational scenarios.

Network-Attached Storage (NAS) Shared Folder Architecture

Governance and Objectives

The purpose of this document is to establish a standardized logical architecture for the organization's central Network-Attached Storage (NAS) system. Unstructured data growth and ad-hoc folder creation inevitably led to operational inefficiencies, data fragmentation, and security vulnerabilities. By defining a rigid directory hierarchy and formal data governance policies from the outset, the organization ensures long-term data integrity, ease of navigation, and auditable access control.

Architectural Design Principles

The storage framework is governed by five core operational principles:

- **Intuitive Taxonomy:** Folder nomenclature relies on clear, unambiguous, and standardized terminology to eliminate operational confusion and misfiling.
- **Principle of Least Privilege (PoLP):** User access is strictly restricted to the minimal data required to fulfill specific job functions. This limits internal data exposure and prevents accidental modification or deletion of critical assets.
- **Standardized Cross-Departmental Templates:** A uniform directory structure is deployed across all business units, fostering administrative consistency and reducing user learning curves.
- **Scalable Architecture:** The directory framework is designed to accommodate organizational expansion, allowing new teams or business units to integrate seamlessly without disrupting the existing hierarchy.
- **Strict Data Isolation:** High-risk departments, specifically Finance and Human Resources, are logically isolated within the directory tree to guarantee strict confidentiality and regulatory compliance.
- **Automated Disaster Recovery:** The entire /Company root directory tree is integrated into an automated nightly backup schedule to ensure business continuity.

Root Directory Topology

To maintain a clean and manageable file system, all organizational data resides beneath a single root directory designated as /Company. The secondary tier of this hierarchy is strictly categorized by department or specific cross-functional business purposes.

/Company

├── Public/ (Global Access – Corporate Distribution)

| ├── Announcements/

| ├── Templates/

| └── Policies/

├── Finance/ (Restricted – Financial Operations)

| ├── Invoices/

| ├── Payroll/

| ├── Budgets/

| └── Reports/

├── HR/ (Restricted – Human Resources)

| ├── Employee-Records/

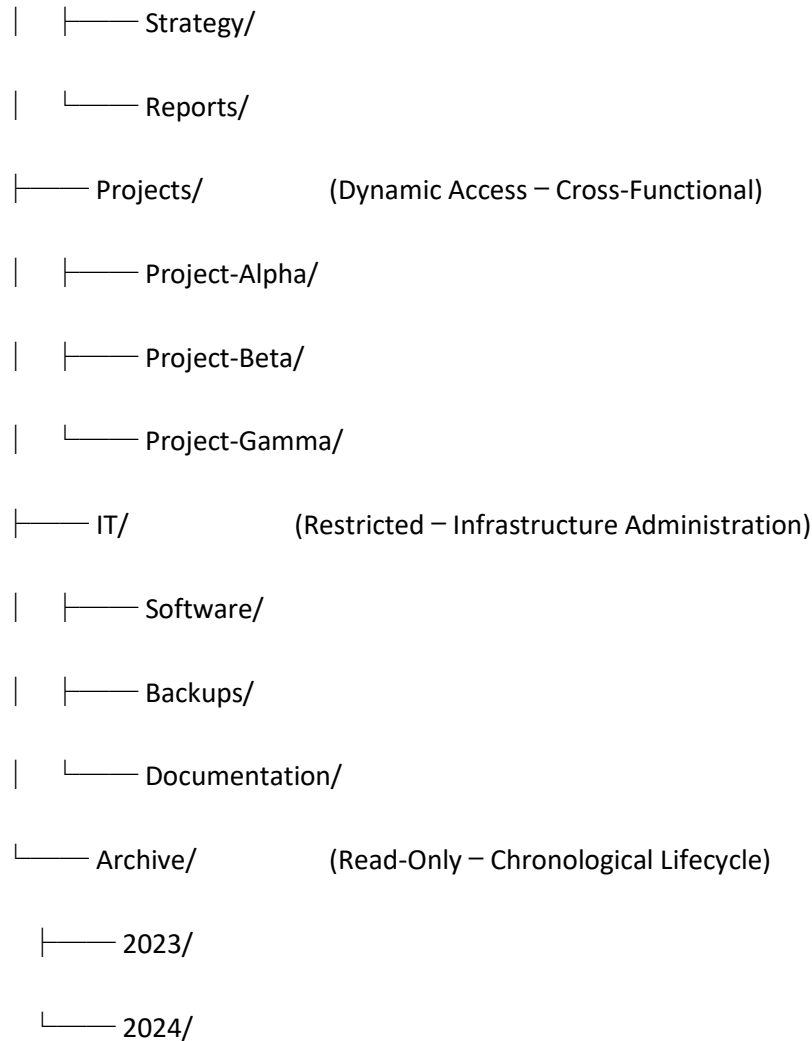
| ├── Recruitment/

| ├── Contracts/

| └── Training/

├── Management/ (Restricted – Executive Leadership)

| ├── Board-Meetings/



Data Repository Taxonomy ("What Lives Where")

To maintain operational security and structural integrity, data is categorized into distinct, functional repositories with tailored access control lists (ACLs):

- **/Company/Public (General Distribution):** This directory serves as the centralized repository for organizational announcements, document templates, and the corporate handbook. It is configured with global write privileges to allow seamless collaboration on non-sensitive corporate assets.

- **/Company/Finance (Financial Operations):** Reserved exclusively for invoices, payroll processing, budgetary frameworks, and fiscal reporting. Access to this directory is strictly audited, logged, and restricted to personnel with authorized financial clearance.
- **/Company/HR (Human Resources):** Contains confidential employee records, employment contracts, and recruitment pipelines. Access is tightly controlled and restricted solely to HR personnel to maintain privacy compliance.
- **/Company/Management (Executive Strategy):** Houses executive board papers, strategic planning documentation, and director-level reporting assets. Access is explicitly restricted to senior management and executive directors.
- **/Company/Projects (Cross-Functional Operations):** Organized dynamically using dedicated subfolders for each active initiative. Because project teams draw members from multiple business units, access permissions are provisioned on a per-project basis rather than by department.
- **/Company/IT (Information Technology):** Reserved for software installer images, systems infrastructure documentation, network configuration mapping, and backup logs. Access is limited strictly to IT administration personnel.
- **/Company/Archive (Data Lifecycle Management):** Formats historical data and completed project files as read-only assets, structured chronologically by calendar year. To preserve data integrity and prevent unauthorized extraction, no assets may be moved out of this directory without formal executive sign-off.

Identity and Access Management (IAM) Protocol

To ensure scalable and maintainable security administration, our team enforces **Role-Based Access Control (RBAC)**.

System permissions are mapped strictly to functional security groups rather than individual user accounts. This architectural standard streamlines employee onboarding and offboarding procedures; administrators only need to modify group memberships rather than auditing or reconfiguring permissions across disparate file systems.

Folder	Read	Write	Full control
/Company/Public	All Staff	All Staff	IT Admins
/Company/Finance	Finance Team	Finance Team	Finance Manager, IT
/Company/HR	HR Team	HR Team	HR Manager, IT
/Company/Management	Managers	Managers	Directors, IT
/Company/Projects	Project Members	Project Members	Project Lead, IT
/Company/IT	IT Team	IT Team	IT Admins
/Company/Archive	Dept Managers	(read-only)	IT Admins

File System Taxonomy and Naming Conventions

To ensure long-term data discoverability and maintain automated scripting compatibility, the organization adheres to a strict file and directory nomenclature standard:

- **Explicit Nomenclature:** Directory and file names must be fully articulated. Abbreviations are prohibited to eliminate ambiguity (e.g., utilizing Employee-Records rather than EmpRec).
- **Delimiter Standardization:** Spaces within file and directory names are prohibited, as they introduce compilation and syntax errors during automated script execution. All spaces must be replaced with hyphens (-).
- **Chronological Ordering Syntax:** All date-dependent files must utilize the ISO 8601 standard (YYYY-MM-DD) as a prefix to ensure optimal linear sorting within file explorers (e.g., 2026-05-27-Invoice-Acme.pdf).
- **Project Directory Formatting:** Active project folders must consistently follow the structural syntax prefix of Project-<Name>.
- **Archival Subdirectory Structure:** Subfolders within the archival directory are to be labeled strictly by the four-digit calendar year (YYYY).

Operational Governance and Data Stewardship

Clearly defined operational roles ensure accountability and protect system resources across three organizational tiers:

- **Information Technology (IT) Department:** Responsible for overall NAS infrastructure management, Active Directory security group maintenance, system updates, and backup integrity.
- **Departmental Managers:** Act as primary data owners. They hold the administrative authority to approve, audit, and modify user memberships within their respective department-level security groups.
- **General End-Users:** Responsible for day-to-day data compliance. Users must ensure assets are saved within their designated directory structures and strictly refrain from placing confidential or sensitive data within the globally accessible Public directory.

Backup Infrastructure and Retention Policy

To mitigate data loss risks and fulfil regulatory compliance requirements, the organization enforces a multi-tiered backup and retention framework:

- **Off-Site Disaster Recovery:** The entire /Company root directory tree undergoes an automated backup sequence to a secure, off-site storage repository on a nightly baseline.
- **Short-Term Snapshot Recovery:** Point-in-time system snapshots are retained for 30 days to facilitate rapid restoration of accidentally modified or deleted files.
- **Long-Term Data Retention:** In accordance with statutory requirements, the /Company/Archive tree is preserved in a permanent, immutable state for a minimum retention period of seven (7) years.

Governance Review and Change Management Schedule

This storage architecture and governance policy is a living document. The IT department, in coordination with departmental business leaders, will conduct a formal structural audit every six (6) months. Out-of-band reviews will be triggered immediately in the event of significant

organizational restructuring, mergers, acquisitions, or any changes that alter the underlying operational assumptions of this framework.

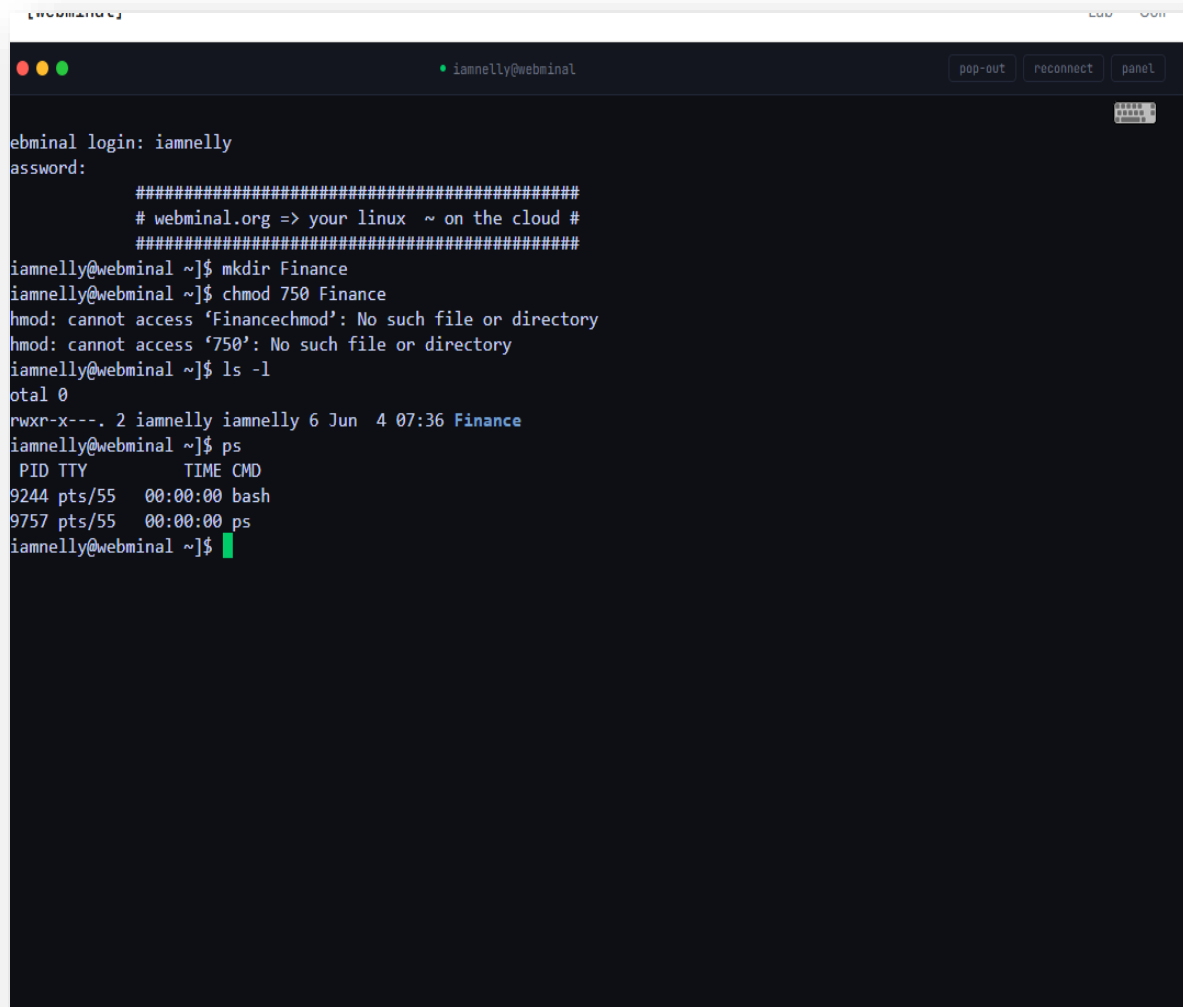
Command-Line Administration

Systems Administration and Directory Security

As part of our infrastructure deployment strategy, our team utilized the Linux command-line interface to establish secure directory structures, manage access control permissions, and audit active system processes.

- **Directory Initialization:** First, we executed the `mkdir Finance` command to provision a dedicated, isolated directory for financial data within the corporate file system.
- **Access Control and Security Compliance:** Next, we hardened the directory's security posture by applying a `chmod 750 Finance` configuration. This specific permission scheme enforces strict role-based access control: it grants the administrator full read, write, and execute privileges (7), allows authorized group members to view and execute resources (5), and completely restricts unauthorized users (0) from accessing sensitive financial assets.
- **System State Verification:** To verify that our security policies were accurately applied, we utilized the `ls -l` command to generate a comprehensive directory listing. As demonstrated in our system logs below, the resulting `drwxr-x---` permission string confirms that the 750 access controls are active, correct, and compliant with data isolation standards.
- **Active Process Auditing:** Finally, we ran the `ps` command to monitor system activity within the current operational session. The generated report verified that only the core command interpreter (`bash`) and the active diagnostic process (`ps`) were running, confirming an optimal and secure execution environment.

Terminal Log Output and Directory Permission Verification



The screenshot shows a terminal window titled "webminal" with a user "iamnelly@webminal". The terminal displays the following commands and output:

```
webminal login: iamnelly
password:
#####
# webminal.org => your linux ~ on the cloud #
#####
iamnelly@webminal ~]$ mkdir Finance
iamnelly@webminal ~]$ chmod 750 Finance
chmod: cannot access 'Financechmod': No such file or directory
chmod: cannot access '750': No such file or directory
iamnelly@webminal ~]$ ls -l
total 0
-rwxr-x---. 2 iamnelly iamnelly 6 Jun  4 07:36 Finance
iamnelly@webminal ~]$ ps
  PID TTY          TIME CMD
 9244 pts/55    00:00:00 bash
 9757 pts/55    00:00:00 ps
iamnelly@webminal ~]$
```

On the right side of the terminal window, there is a sidebar with the following elements:

- A header "IA iamnelly connected" with a "Lessons" button.
- A section "// SELECT LESSON" with a "Select" button.
- A message "Select a lesson" and "Content will appear".
- A "prev" button at the bottom.

Software Deployment Plan

Endpoint Management and Deployment Strategy

To maximize operational efficiency for our fleet of 25 workstations, our team will utilize centralized deployment rather than manual, machine-by-machine installations. Manual configuration is inefficient and fails to scale alongside organizational growth.

Because an Active Directory (AD) environment is already established, we will leverage native **Group Policy Objects (GPOs)** for centralized software and policy distribution. While enterprise platforms like Microsoft Endpoint Configuration Manager (MECM) or cloud-based Microsoft Intune would be considered for larger or hybrid fleets, GPOs provide a robust, zero-cost solution for an organization of our current scale.

Our deployment framework includes the following baseline initiatives:

- **Standardized Operating System Images:** We will develop a master Windows image pre-configured with essential productivity software (Microsoft Office), endpoint security (antivirus), and standard web browsers. New hardware deployment will be streamlined via Preboot Execution Environment (PXE) network booting or standardized USB cloning.
- **Application Repackaging:** Software installers that do not natively support standard installation formats will be repackaged into .msi or .msix formats. This ensures all deployments execute silently in the background without user intervention.
- **Automated Patch Management:** Software updates and system pushes will be scheduled outside of standard business hours. Enforcing automated update schedules prevents configuration drift and ensures consistent security compliance across the network.

Ultimately, utilizing existing GPO infrastructure eliminates unnecessary procurement costs and avoids the operational overhead of managing redundant third-party administrative tools.

IP Addressing Scheme and Subnetting

The historical reliance on a flat network architecture introduces significant broadcast traffic risks and scaling bottlenecks as the device count increases. To mitigate this, the network has been segmented into distinct /24 subnets under the private 192.168.0.0/16 address space.

Allocating dedicated subnets based strictly on device functionality drastically simplifies firewall policy enforcement, streamlines network administration, and isolates broadcast domains to improve overall network performance.

Subnet	Range	What lives there
Servers & infrastructure	192.168.10.0/24	Gateway, DNS, DHCP, file server, domain controller
User workstations	192.168.20.0/24	All 25 desktops and laptops
Printers & peripherals	192.168.30.0/24	Network printers, scanners, UPS management
Wi-Fi / guest	192.168.40.0/24	Wireless clients and the isolated guest VLAN
Management / IoT	192.168.50.0/24	IP cameras, door access, HVAC controllers

Network Segmentation and Access Control

Implementing network segmentation ensures that broadcast domains are isolated, effectively optimizing traffic flow and reducing overall network noise. Furthermore, this architectural approach simplifies the implementation of firewall policies; rather than managing a complex list of security exceptions, a single global rule can be deployed to restrict guest networks from communicating with production servers.

Additionally, establishing a structured IP addressing scheme enhances troubleshooting efficiency. By assigning distinct IP subnets to specific categories of equipment, administrators can

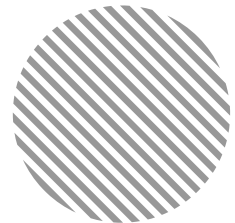
immediately identify a device's functional role based solely on its network address, significantly reducing the time required for system diagnostics.

Conclusion & Outlook

The structural configurations detailed within this report establish a verified, secure, and highly manageable operational baseline for **Ubuntu Innovations (Pty) Ltd.**'s current fleet of 25 workstations and centralized data repositories. By leveraging native Active Directory environments and Group Policy Objects, the organization achieves enterprise-grade endpoint configuration management and patch automation without incurring unnecessary software procurement costs. Simultaneously, the newly deployed Network-Attached Storage (NAS) architecture enforces strict data isolation for high-risk assets, standardizes file naming conventions, and ensures automated disaster recovery resilience.

Moving forward, the systematic six-month governance review schedule will ensure that these access control frameworks, IP addressing subnets, and storage taxonomies adapt dynamically alongside evolving business operations. Ultimately, this infrastructure deployment successfully mitigates security risks, eliminates administrative bottlenecks, and guarantees structural data compliance across the entire enterprise of **Ubuntu Innovations (Pty) Ltd.**

**Ubuntu Innovations
(Pty) Ltd**



**System Administration &
Cybersecurity**



Prepared by :
Khusela Ikamva Tech

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1. Introduction

Ubuntu Innovation (Pty) Ltd operates in an increasingly digital environment where the confidentiality, integrity and availability of company information are central to our ability to serve clients, retain trust and remain competitive. As a team, we recognise that our data, systems, intellectual property and the personal information of our employees and customers are among our most valuable assets. The loss, theft or compromise of any of these would have a direct impact on our operations, reputation, regulatory standing and bottom line.

This report has been prepared collaboratively by the IT, Risk and Operations team as part of the company's "Protect the Company" initiative. Our brief was clear: design a practical, end-to-end strategy that strengthens how we back up our data, secure our systems and respond when something goes wrong. The intention is not to introduce technology for its own sake, but to ensure that every reasonable step has been taken to safeguard the business against the threats most likely to affect a South African mid-sized technology company in 2026.

1.1 Purpose of the Report

The purpose of this report is to consolidate, in a single working document, the policies, plans and procedures that together form Ubuntu Innovation's information security and business continuity posture. It is meant to serve three audiences at once: executive management, who require a clear summary of risks and controls; line managers, who need actionable procedures for their teams; and operational staff, who must understand their day-to-day responsibilities.

1.2 Scope

This report covers five interconnected areas, each of which has been developed as a stand-alone section that can also be read in isolation:

1. A Backup and Disaster Recovery Plan covering data classification, backup schedules, storage locations, retention and recovery testing.
2. A Cybersecurity Policy setting out the technical and behavioural controls expected of all employees, contractors and third parties.
3. A Risk Assessment Matrix identifying the most material information security risks, with likelihood, impact and mitigation actions.
4. An Incident Response Plan describing how we detect, contain, eradicate, recover from and learn from security incidents.
5. A Security Awareness Training Outline designed to build a security-conscious culture across every department.

1.3 Approach

In preparing this report we reviewed our current IT environment, interviewed staff across departments, benchmarked against widely accepted frameworks such as ISO/IEC 27001 and the NIST Cybersecurity Framework, and considered our obligations under the Protection of Personal Information Act (POPIA). Where existing controls are working, we have documented them; where gaps were identified, we have proposed specific, costed and time-bound improvements.

1.4 Key Assumptions

- All Ubuntu Innovation staff use company-issued devices for company work and bring-your-own-device usage is limited and managed.
- The company maintains a primary office network, a small server room and a mix of cloud-hosted services (Microsoft 365, customer CRM, code repositories).
- Internet connectivity is provided by a single primary fibre line with an LTE failover.
- Management has indicated willingness to fund security investments that are clearly justified by risk reduction.

2. Backup and Disaster Recovery Plan

Our backup and disaster recovery (DR) plan exists to ensure that, no matter what happens — accidental deletion, hardware failure, ransomware, fire, flood or theft — Ubuntu Innovation can recover its critical data and resume operations within a defined and acceptable timeframe. This section sets out what we back up, where we store it, how often, how long we keep it and how we prove that the backups work.

2.1 Objectives

- **Recovery Time Objective (RTO):** critical systems must be restored within 4 hours; non-critical systems within 24 hours.
- **Recovery Point Objective (RPO):** no more than 24 hours of data loss for any critical system, and no more than 1 hour for transactional databases.
- **Resilience:** no single event - including the total loss of the primary office — should be capable of permanently destroying company data.
- **Compliance:** backup practices must satisfy POPIA requirements regarding the safeguarding and retention of personal information.

2.2 Data Classification

Not all data carries the same value or sensitivity. We classify company data into four tiers, and our backup frequency and retention are determined by this classification:

Tier	Examples	Backup Frequency	Retention
Critical	Customer database, financial records, source code, signed contracts	Continuous / hourly	7 years
Important	Email, shared drives, project files, HR records	Daily	3 years
Operational	Workstation user profiles, internal wikis, meeting notes	Weekly	12 months
Transient	Cache, temp files, downloaded media	Not backed up	N/A

2.3 The 3-2-1-1 Backup Rule

Ubuntu Innovation applies the industry-standard 3-2-1-1 rule. This means that at any given moment we maintain at least three copies of every critical data set, on two different types of media, with one copy stored off-site and one copy held in immutable (write-once) storage that cannot be altered or deleted by ransomware.

- **Copy 1 - Production:** the live data on our servers and cloud platforms.

- **Copy 2 - Local NAS:** an on-premises Network-Attached Storage device in the server room, receiving incremental backups every night.
- **Copy 3 - Cloud:** encrypted weekly full backups replicated to a reputable cloud provider in a geographically separate region.
- **Copy 4 - Immutable archive:** monthly snapshots written to object-lock storage that cannot be modified for 90 days, protecting against ransomware that targets backups themselves.

2.4 Backup Schedule

Backup Type	Frequency	Destination	Owner
Database transaction logs	Every 60 minutes	Local NAS + cloud	Database Administrator
Full file server backup	Daily, 20:00	Local NAS	IT Support Lead
Microsoft 365 (mail, OneDrive, SharePoint)	Daily	Third-party SaaS backup	IT Support Lead
Full server image	Weekly, Sunday 22:00	Encrypted cloud	Systems Administrator
Immutable archive snapshot	Monthly, 1st of month	Object-lock cloud storage	IT Manager

2.5 Encryption and Security of Backups

All backup data is encrypted in transit using TLS 1.2 or higher, and at rest using AES-256. Encryption keys are managed in a dedicated key vault, with access restricted to two named senior administrators. Backup media taken off-site for any reason are logged in and out of the server room by the duty IT officer.

2.6 Disaster Recovery Procedures

In the event of a disaster, the following sequence is followed. The IT Manager (or delegate) is the designated Recovery Coordinator and has the authority to declare a disaster and initiate the DR plan.

6. Declare the incident: confirm scope, notify executive management and activate the incident response team.
7. Assess: determine which systems are affected and whether the primary site is usable.
8. Failover: redirect critical services (email, CRM, customer portal) to their cloud-hosted DR equivalents.
9. Restore: rebuild affected servers from the most recent clean backup, working from highest to lowest tier.
10. Verify: confirm data integrity, application functionality and that no malware persists.
11. Communicate: keep staff, customers and, where required, regulators informed throughout.

-
12. Return to normal: once the primary environment is stable and verified, fail services back during a planned maintenance window.

2.7 Testing and Assurance

A backup is only as good as the last successful restore. We therefore perform tiered testing on a defined cadence:

- **Monthly:** random-sample file restores from the local NAS, performed by the IT Support Lead and logged.
- **Quarterly:** full database restores to an isolated test environment, verified by the Database Administrator.
- **Annually:** a full disaster recovery simulation covering total loss of the primary site, including measurement of actual RTO and RPO against the targets above.

Results of each test are recorded in the Backup Test Register and reviewed by the IT Manager. Any failures trigger a corrective action with a 14-day remediation deadline.

3. Cybersecurity Policy

This policy sets out the minimum standards of behaviour and the technical controls that protect Ubuntu Innovation's information assets. It applies to every employee, contractor, intern and third-party service provider who accesses our systems or data. By using a company device or account, individuals agree to abide by this policy. Non-compliance may result in disciplinary action up to and including dismissal, and in serious cases criminal or civil proceedings.

3.1 Acceptable Use

- Company devices and accounts are provided for business purposes; limited personal use is permitted provided it is lawful, does not interfere with work and does not consume excessive resources.
- Users may not install unauthorised software, disable security tools, or connect personal storage devices without IT approval.
- Confidential information must never be sent to personal email accounts or uploaded to personal cloud services.

3.2 Identity and Access Management

- **Unique accounts:** every user has a unique, named account; shared logins are prohibited.
- **Least privilege:** users receive only the access they need to perform their role, reviewed quarterly by line managers.
- **Privileged access:** administrator accounts are separate from day-to-day accounts and require additional approval.
- **Joiners, movers, leavers:** HR notifies IT of all role changes; access is granted, modified or revoked within one business day.

3.3 Password and Authentication Standards

- Passwords must be at least 14 characters long, ideally a passphrase, and must not be reused across systems.
- Multi-Factor Authentication (MFA) is mandatory for email, VPN, administrator accounts and any cloud service holding company data.
- Passwords are stored only in the company-approved password manager; writing passwords on paper or in unencrypted files is prohibited.
- Default passwords on any new device or system must be changed before the device is placed into service.

3.4 Device and Endpoint Security

- All laptops and desktops run the company-managed endpoint detection and response (EDR) agent, with tamper protection enabled.
- Full-disk encryption (BitLocker for Windows, FileVault for macOS) is enforced on every portable device.
- Operating system and application security updates are applied automatically within 14 days of release; critical patches within 72 hours.
- Screens must lock automatically after 5 minutes of inactivity, and users must lock their screens manually when leaving a workstation.

3.5 Network Security

- The corporate Wi-Fi is segmented from a separate guest network; guests never access the corporate LAN.
- Remote access to internal systems requires the company VPN with MFA.
- Inbound firewall rules follow a default-deny posture; exceptions are documented and reviewed quarterly.
- Intrusion detection logs are reviewed daily by the IT Support Lead and weekly by the IT Manager.

3.6 Data Protection and Handling

- Personal information processed by Ubuntu Innovation is handled in accordance with POPIA, including the lawful basis for processing and the data subject's rights.
- Confidential documents must be classified using the four-tier scheme set out in Section 2.2 and labelled accordingly.
- Sensitive data shared externally must be encrypted (password-protected archives, secure file transfer, or encrypted email).
- Printed confidential material must be collected from printers promptly and shredded when no longer required.

3.7 Email, Web and Messaging

- All inbound and outbound email passes through an anti-spam and anti-phishing gateway with attachment sandboxing.
- Users must not click on suspicious links or open unexpected attachments; suspected phishing should be reported via the 'Report Phishing' button in Outlook.
- Use of personal messaging apps for company business is discouraged; where unavoidable, no confidential data may be shared.

3.8 Third-Party and Vendor Management

- Any new vendor with access to company data must complete a security questionnaire and sign a Data Processing Agreement.
- Vendor access is time-bound, logged and reviewed every six months.
- Critical vendors must demonstrate ISO 27001 certification or an equivalent recognised standard.

3.9 Policy Governance

This policy is owned by the IT Manager and approved by the Executive Committee. It is reviewed annually, or sooner if there is a material change in the threat landscape, regulatory environment or company operations. All staff acknowledge the policy on joining and at each annual refresh.

4. Risk Assessment Matrix

To prioritise our security investments, the team conducted a structured risk assessment. Each identified risk was scored on likelihood (1–5) and impact (1–5); the product of the two gives a risk rating from 1 (negligible) to 25 (critical). The matrix below summarises the top risks currently facing Ubuntu Innovation, together with the existing controls and the additional mitigation actions we recommend.

4.1 Risk Scoring Key

- 1–5 (Low): accept and monitor.
- 6–12 (Medium): mitigate with reasonable controls; review quarterly.
- 13–19 (High): mitigate as a priority within 90 days; report to executive.
- 20–25 (Critical): immediate executive attention and remediation.

4.2 Top Risks

Risk	Likelihood	Impact	Rating	Mitigation
Phishing / business email compromise	5	4	20 — Critical	Advanced email filtering with attachment sandboxing; mandatory quarterly phishing simulations; MFA on all accounts; finance dual-approval rule for any payment instruction received by email.
Ransomware	3	5	15 — High	Immutable backups; EDR with behavioural detection; network segmentation; least-privilege access; tested DR plan with quarterly restore drills.
Insider data theft	2	5	10 — Medium	Data Loss Prevention on email and USB; activity logging on confidential repositories; exit checklist with same-day revocation; mandatory leave for sensitive roles.
Lost or stolen laptop	4	3	12 — Medium	Full-disk encryption; remote wipe via mobile device management; asset register; SAPS case opened for every theft.
Cloud service misconfiguration	3	4	12 — Medium	Cloud Security Posture Management tooling; baseline secure configuration; quarterly configuration review; change control for all tenant-level settings.
Power outage / load shedding	5	2	10 — Medium	UPS on all servers and network equipment; generator with weekly test run; documented graceful shutdown procedure.
Third-party / supply chain breach	3	4	12 — Medium	Vendor risk questionnaire; contractual security obligations; restricted vendor access; monitoring of vendor security bulletins.

Risk	Likelihood	Impact	Rating	Mitigation
DDoS attack on customer portal	2	4	8 — Medium	Cloud-based DDoS protection in front of public services; rate limiting; runbook for traffic surge events.
POPIA non-compliance	3	4	12 — Medium	Appointed Information Officer; data inventory; consent and retention controls; breach notification procedure aligned to Section 22.

4.3 Risk Treatment

For each risk rated Medium or higher, an action owner and target completion date has been assigned in the Risk Register, which is reviewed monthly by the IT Manager and quarterly by the Executive Committee. Residual risks that fall within the Board's stated risk appetite are formally accepted and documented.

5. Incident Response Plan

Despite the controls described in this report, security incidents remain a matter of when, not if. The Incident Response Plan (IRP) ensures that when one occurs, Ubuntu Innovation responds in a calm, structured and effective way that minimises damage, preserves evidence and restores normal operations as quickly as possible.

5.1 The Incident Response Team (IRT)

- **Incident Commander - IT Manager:** owns the response, makes containment decisions and authorises external communication.
- **Technical Lead - Systems Administrator:** coordinates investigation, containment and recovery activities.
- **Communications Lead - Marketing Manager:** drafts internal and external messaging under guidance from the Incident Commander.
- **Legal & Compliance Lead - Company Secretary:** advises on regulatory obligations, including POPIA breach notifications.
- **HR Lead - HR Manager:** manages any employee-related aspects, including suspected insider incidents.

5.2 Incident Severity Levels

Level	Definition	Response Target
SEV-1 Critical	Active ransomware, confirmed data breach, full system outage	IRT mobilised within 15 minutes; executive notified within 30 minutes
SEV-2 High	Targeted phishing campaign, single-user account compromise, partial outage	IRT mobilised within 1 hour; resolved within 24 hours
SEV-3 Medium	Malware detected and contained by EDR, lost laptop, suspicious activity	Investigated within 1 business day
SEV-4 Low	Policy violation, spam reaching inbox, minor misconfiguration	Logged and addressed in normal IT workflow

5.3 Response Lifecycle

Our response follows the six-phase NIST lifecycle, adapted to our environment:

Phase 1 — Preparation

- Maintain up-to-date contact lists for the IRT, executives, vendors and law enforcement (SAPS Cybercrime Unit, Hawks).

-
- Pre-stage forensic tools, clean laptops and out-of-band communication (a private WhatsApp group hosted on personal devices for use when corporate email is unavailable).
 - Maintain a retainer with an external incident response firm for surge capacity.

Phase 2 — Detection and Analysis

- Alerts originate from EDR, email gateway, firewall, user reports or external notification.
- The on-call IT engineer triages the alert within 15 minutes and assigns a severity level.
- All actions are logged in the Incident Log from the moment the incident is declared.

Phase 3 — Containment

- Short-term: isolate affected devices from the network (disconnect cable, disable Wi-Fi, quarantine via EDR).
- Account-level: force password resets, revoke active sessions, disable compromised accounts.
- Long-term: apply firewall rules, block malicious indicators across the estate.

Phase 4 — Eradication

- Identify and remove the root cause: malicious files, persistence mechanisms, attacker accounts.
- Patch the exploited vulnerability before bringing systems back online.
- Where compromise is suspected at OS level, reimage rather than clean.

Phase 5 — Recovery

- Restore from a verified clean backup taken before the incident window.
- Monitor restored systems intensively for 14 days for signs of re-infection.
- Phased return to service, lowest-risk systems first.

Phase 6 — Lessons Learned

- Within 10 business days the IRT holds a blameless post-incident review.
- Findings are written up in a Post-Incident Report including timeline, root cause, impact, what worked, what did not, and concrete improvement actions with owners and dates.
- The Risk Register and this report are updated where necessary.

5.4 Worked Example — Ransomware Detected on a Finance Workstation

13. EDR alert at 09:14: 'Ransom note created by suspicious process' on workstation FIN-07. On-call engineer acknowledges within 5 minutes.
14. Severity declared SEV-1. Incident Commander activates the IRT and notifies the CEO and CFO.
15. Containment: FIN-07 isolated via EDR; the entire finance VLAN is segmented from the rest of the network; the user's account is disabled and all active sessions revoked.

-
16. Scope assessment: log review identifies one mapped share that the workstation could write to; that file server is taken offline as a precaution; immutable backups are confirmed intact.
 17. Eradication: FIN-07 is reimaged from a known-good gold image; the patch for the exploited vulnerability (a malicious macro in a Word document) is rolled out to all endpoints; macro execution from internet-sourced documents is blocked via Group Policy.
 18. Recovery: the affected file server is restored from the previous night's backup; checksums verified; users notified that any work done between 17:00 yesterday and the incident may need to be re-entered.
 19. Communications: an internal note is issued to all staff explaining the incident in plain language and reminding them of phishing reporting procedures.
 20. Regulatory: the Information Officer confirms that no personal information was exfiltrated; a precautionary record is kept in case POPIA notification becomes necessary.
 21. Lessons learned (held 7 days later): macro policy gap closed; phishing simulation focused on macro lures scheduled; backup immutability validated as the single most important control in limiting damage.

6. Security Awareness Training Outline

Technology alone cannot keep Ubuntu Innovation safe. The overwhelming majority of successful attacks begin with a person clicking a link, approving a fraudulent payment, plugging in an unknown USB stick. A workforce that is alert, sceptical and confident in reporting suspicious activity is the single most cost-effective security control available to us. This section describes how we will build and sustain that culture.

6.1 Training Objectives

- Every employee understands the most common threats relevant to their role and can recognise them in practice.
- Every employee knows what to do - and what not to do when something looks suspicious.
- Reporting a suspected incident is seen as a positive contribution, never as a fault to be hidden.
- Security expectations are reinforced often enough to remain front-of-mind, without becoming background noise.

6.2 Training Programme Structure

Component	Audience	Frequency	Format
Induction session	New joiners	Within first week	60-minute facilitated session + policy sign-off
Annual refresher	All staff	Once per year	E-learning module + short assessment
Phishing simulations	All staff	Quarterly	Realistic test emails; instant micro-learning for those who click
Role-based deep dives	Finance, HR, IT, Developers	Twice yearly	Workshop tailored to threats specific to the role
Security moments	All meetings	Ongoing	3-minute slot at the start of departmental meetings
Executive briefing	Executive Committee	Quarterly	Threat landscape, incident trends, key metrics

6.3 Sample 30-Minute Induction Session Agenda

- **00:00 – 00:03 Welcome and why this matters:** real South African case studies of companies that were materially harmed by a single click.
- **00:03 – 00:10 Spotting phishing:** sender mismatch, urgency cues, suspicious links, attachments — live demonstration of a real phish.
- **00:10 – 00:16 Password and account hygiene:** passphrases, the company password manager, MFA enrolment walkthrough.

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- **00:16 – 00:21 Data handling:** classification scheme, sharing rules, what never leaves the company.
 - **00:21 – 00:26 If something goes wrong:** how to report, who to contact, the 'no blame for honest mistakes' principle.
 - **00:26 – 00:30 Questions and policy acknowledgement signing.**

6.4 Measuring Effectiveness

- Phishing simulation click rate, tracked over time and by department.
- Time-to-report for simulated phishing emails - we want this trending down.
- Completion rate of annual e-learning, target 100% within the cycle.
- Number of voluntarily reported suspicious emails per month — higher is better, indicating a vigilant culture.
- Reduction in real incidents traceable to user behaviour, year on year.

6.5 Reinforcement and Culture

Awareness is built through repetition in small doses rather than long annual lectures. We will use short monthly bulletins, posters in common areas, and a dedicated channel on the company intranet to keep security visible. Managers will be expected to model the behaviours expected of their teams. Recognition for staff who report genuine threats will be built into the existing employee recognition programme.

7. Conclusion

Protecting Ubuntu Innovation (Pty) Ltd is not a one-off project but an ongoing discipline. The five pillars set out in this report - disciplined backups, a clear cybersecurity policy, an honest risk assessment, a tested incident response plan and continuous awareness training work together as a single, coherent programme. No one of them alone is sufficient; together they provide defence in depth that significantly reduces both the likelihood and the impact of the security events most likely to affect us.

As a team we are realistic. We know that determined attackers will continue to evolve their methods, that load shedding and connectivity issues will continue to test our resilience, and that human error will always be part of the equation. The objective is not to achieve a state of perfect security, which does not exist, but to ensure that when something goes wrong, we recover quickly, learn openly, and emerge stronger.

We are also pragmatic. The recommendations in this report have been kept proportionate to the size, risk profile and resources of Ubuntu Innovation. Where investment is required — for example in immutable backup storage, EDR licences, or external incident response retainers — the cost has been weighed against the potential cost of the event it prevents. In every case the business case is clear.